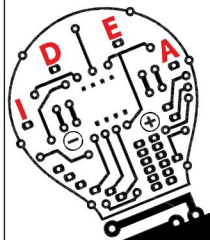


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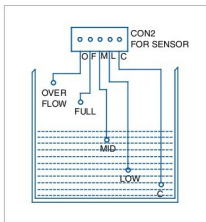


Fig. 7: Sensors in the overhead tank



Fig. 8: Author's prototype of the transmitter unit



Fig. 9: Author's prototype of the receiver unit
with LED display

in separate plastic boxes.

You can use 9V regulated power supplies instead of batteries to power the circuits. For troubleshooting, check voltages at various points mentioned in the test point table.

Make a separate PCB for the LED indication (shown by the dotted

PARTS LIST

Semiconductors:

- IC1, IC3 - 7805, 5V voltage regulator
- IC2 - HT12E encoder
- IC4 - HT12D decoder
- T1-T4 - BC547 npn transistor
- LED1-LED7 - 5mm LED

Resistors (all 1/4-watt, 5% carbon):

- R1, R11, R12 - 470-ohm
- R2 - 1-mega-ohm
- R3-R6, R14-R17 - 10-kilo-ohm
- R7-R10 - 2.2-kilo-ohm
- R13 - 47-kilo-ohm
- R18-R21 - 330-ohm

Capacitors:

- C1, C3 - 100µF, 25V electrolytic
- C2 - 10µF, 16V electrolytic
- C4 - 100nF ceramic disk

Miscellaneous:

- DIP1, DIP2 - 8-way DIP switch
- TX1 - 433MHz transmitter module
- RX1 - 433MHz receiver module
- CON1, CON3 - 2-pin terminal connector
- CON2 - 5-pin connector
- CON4 - 5-pin female burg strip
- CON5 - 5-pin male burg strip
- ANT.1, ANT.2 - 10-15cm-long wire antenna
- 9V regulated power supply / 9V battery (2 numbers)

lines on the right side of Fig. 6). Use different LED colours for different levels.

Use a waterproof box for the transmitter unit to protect it from rain water.

When both the circuits are switched on and LED3 glows, it indicates that the transmitter unit and the receiver unit are connected successfully. If the addresses of the units are different, LED3 will not glow.

Use DIP switches (DIP1 and DIP2) to set the addresses. Switch on pins 1, 2 and 7 of both the DIP switches to make a secure connection. This will prevent interference from nearby RF transmitters. Connect the sensors in the tank as shown in Fig. 7.

Author's prototypes of the transmitter and receiver units are shown in Figs 8 and 9, respectively. **EFY**



Pranav Rameshji Rokde is a diploma (electronics engineering) student at Shri Datta Meghe Polytechnic, Nagpur, Maharashtra